

Mean stress effects in stress-life fatigue and the Walker equation

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ABSTRACT Mean stress effects in finite-life fatigue are studied for a number of sets of experimental data for steels, aluminium alloys and one titanium alloy. Specifically, the agreement with these data is examined for the Goodman, Morrow, Smith–Watson–Topper and Walker equations. The Goodman relationship is found to be highly inaccurate. Reasonable accuracy is provided by the Morrow and by the Smith–Watson–Topper equations. But the Morrow method should not be used for aluminium alloys unless the true fracture strength is employed, instead of the more usual use of the stress-life intercept constant. The Walker equation with its adjustable fitting parameter γ gives superior results. For steels, γ is found to correlate with the ultimate tensile strength, and a linear relationship permits γ to be estimated for cases where non-zero mean stress data are not available. Relatively high-strength aluminium alloys have $\gamma \approx 0.5$, which corresponds with the SWT method, but higher values of γ apply for relatively low-strength aluminium alloys. For both steels and aluminium alloys, there is a trend of decreasing γ with increasing strength, indicating an increasing sensitivity to mean stress.

Keywords fatigue of materials; Goodman equation; mean stress effect; Morrow equation; Smith–Watson–Topper equation; stress-life curve; Walker equation.

NOMENCLATURE

A = intercept constant at 1 cycle for a stress-life curve
 b = exponent constant for a stress-life curve
 b_w = exponent constant for a Walker method stress-life fit
 c = exponent constant for a plastic strain versus life curve
 d = intercept constant for multiple linear regression
 E = elastic modulus
 m_1, m_2 = slope constants for multiple linear regression
 n = number of data points for an s_z calculation
 N^* = life for a given ε_a for the $\sigma_m = 0$ case
 N_w^* = value of N^* from the Walker method
 N_f = fatigue life; cycles to failure
 R = stress ratio, $R = \sigma_{\min}/\sigma_{\max}$
 s_z = stress deviation; the standard deviation of z for a set of data
 z = normalized stress-direction deviation of a data point relative to a stress-life curve
 $\Delta\sigma$ = stress range, $\Delta\sigma = 2\sigma_a$
 ε_a = strain amplitude
 ε_{ar} = strain amplitude for $\sigma_m = 0$
 ε'_f = intercept constant at 1/2 cycle for a plastic strain versus life curve

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- γ = fitting constant for the Walker method
 σ_a = stress amplitude
 σ_{ar} = stress amplitude for $\sigma_m = 0$; equivalent completely reversed stress amplitude
 σ_{arN} = point on a σ_{ar} versus N_f curve at the same N_f as σ_{arP}
 σ_{arP} = value of σ_{ar} for the stresses applied in a test
 σ'_f = intercept constant at 1/2 cycle for a stress-life curve
 σ'_{fw} = intercept constant at 1/2 cycle for a Walker method stress-life fit
 $\tilde{\sigma}_{fB}$ = true fracture strength, corrected
 σ_m = mean stress
 σ_{max} = maximum stress
 σ_{min} = minimum stress
 σ_u = ultimate tensile strength

INTRODUCTION

The effect of mean stress in fatigue of materials has been a subject of study for well over a century. Numerous methods have been proposed and employed for handling mean stresses in the calculation of allowable cyclic stresses and fatigue lives. However, despite all of this work, there is no general consensus as to the suitability of various approaches, and methods of questionable accuracy continue to be recommended in textbooks and given as options in computer programs.

This paper attempts to aid the situation by reporting an extensive study of existing fatigue data that include mean stress effects. The analysed data are limited to engineering metals and to axially loaded, unnotched specimens. Included are a total of 28 sets of data, with 18 of these being for steels, nine for aluminium alloys and one for a titanium alloy. Each set of data is compared to the behaviour expected from the modified Goodman, Morrow, Smith–Watson–Topper and Walker methods. Comparisons are made in the context of stress-life curves in the finite-fatigue-life region. Fatigue limits, that is, stress levels below which fatigue damage is presumed not to occur, are not specifically included in the study.

The work reported here follows a preliminary assessment of eight sets of data by one of the authors.¹ In that paper, the Walker method was found to be worthy of special attention, which emphasis in continued here.

The nomenclature employed is defined in Fig. 1, where σ_a is stress amplitude, σ_m is mean stress, and $\Delta\sigma = 2\sigma_a$ is the stress range. The stress ratio $R = \sigma_{min}/\sigma_{max}$ is also employed. It is useful to note some relationships among the quantities used to describe cyclic loading.

$$\begin{aligned}
 \sigma_a &= \frac{\sigma_{max} - \sigma_{min}}{2} & (a) \\
 \sigma_m &= \frac{\sigma_{max} + \sigma_{min}}{2} & (b) \\
 \sigma_a &= \frac{\sigma_{max}}{2}(1 - R) & (c) \\
 \sigma_m &= \frac{\sigma_{max}}{2}(1 + R) & (d)
 \end{aligned}
 \tag{1}$$

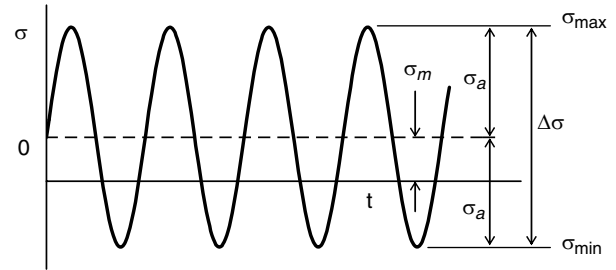


Fig. 1 Constant amplitude cyclic stressing and definitions of stress variables.

The data analysed are identified in Table 1 as to the material, literature references,^{2–15} and tensile and other properties. Some of the materials are listed with the AISI–SAE or Aluminium Association designations which are prevalent in the United States, and others with German nomenclature from the handbooks of Baumel and Seeger⁴ and of Boller and Seeger.¹²

MEAN STRESS RELATIONSHIPS

As a basis for comparison, it is assumed that each material has a stress versus fatigue life relationship for completely reversed (zero mean stress) loading of the following standard form:

$$\sigma_{ar} = \sigma'_f (2N_f)^b \tag{2}$$

The quantities σ'_f and b are fitting constants, N_f is cycles to failure and σ_{ar} is the stress amplitude. The subscript r is added to σ_a to indicate the $\sigma_m = 0$ special case. This power relationship of course gives a straight line on a log–log plot, as shown for two cases in Fig. 2.

Constants for Eq. (2), as fitted in this study to the $\sigma_m = 0$ data in each set, are given in Table 1. A reasonable fit over the entire range of the data was generally possible. There were three exceptions, all aluminium alloys, where the data extended to very short lives. In these cases, the data at short lives fell well below the trend of the data at longer

Table 1 Materials, tensile properties and stress-life fitting constants

Material ^a	Source ^b	Yield σ_y (MPa)	Ultimate σ_u (MPa)	Fracture $\bar{\sigma}_B$ (MPa)	Red. Area ^g (Elong.) (%)	Fit $\sigma_m = 0$		Fit all σ_m to Walker Equation		Range of Walker Fit		
						σ'_f (MPa)	b	σ'_{fu} (MPa)	b_w	γ	Min (cycles)	Max (cycles)
SAE 1015 St	2, 3	228	415	726	67.9	1020	-0.1379	990	-0.1336	0.7352	52	1,241,000
GSMnNi63 St ^{c,d}	4	312	501	846	(26)	672	-0.0811	739	-0.0925	0.8113	15	1,900,000
CK45 St	4	531	790	1271	60	1267	-0.0984	1217	-0.0936	0.6949	217	620,000
SAE 4130 St, Norm ^d	5, 6	647	799	1144	(15.2)	1168	-0.0845	1584	-0.1040	0.6903	211	1,984,000
49MnVS3 St	4	566	840	1152	44	1325	-0.0948	1263	-0.0899	0.8492	246	1,710,000
17MnCrMo33 St	4	833	929	1446	58	1232	-0.0666	1275	-0.0752	0.6575	23	2,840,000
CC 450 SS, H1150 ^d	7	678	1015	1360	(23.2)	1365	-0.0591	1453	-0.0668	0.6253	157	72,950
CC 450 SS, Soln An ^d	7	1041	1065	1410	(12.2)	1326	-0.0600	2046	-0.1078	0.7051	231	94,613
50CrMo4 St, El Pol	4	970	1086	1609	48.6	1547	-0.0841	1588	-0.0899	0.7776	52	380,000
42CrMo4 St	4	998	1111	1525	60	1452	-0.0758	1505	-0.0810	0.7781	248	520,000
AISI 4340 St ^c	8	1103	1172	1634	56	1722	-0.0958	1963	-0.1077	0.6497	222	901,430
SAE 4130 St, Hard ^d	5, 6	1200	1241	1586	(8.3)	2437	-0.1182	2378	-0.1160	0.5457	74	1,023,000
CC 450 SS, H900 ^d	7	1354	1405	1750	(15.3)	2317	-0.0840	2839	-0.1083	0.4758	223	90,941
PH13-8Mo SS, H1000 ^d	9	1358	1413	1758	-	2889	-0.0995	2521	-0.0879	0.5769	2797	1,266,000
300M St ^d	9	1634	1958	2303	-	3612	-0.1150	4214	-0.1318	0.4157	684	652,600
SAE 1045 St, 705 HB	10	1827	2082	2131	2	2418	-0.0750	2370	-0.0728	0.4839	2.5	161,250
SAE 1045 St, 55 HRC	11	1731	2165	2690	38	3542	-0.1075	3762	-0.1147	0.4286	26	750,791
SAE 1045 St, 595 HB	10	1841	2248	2717	40.5	3149	-0.1014	3034	-0.0985	0.5245	20	572,000
6061-T6 Al ^e	9	276	310	420	-	722	-0.1151	651	-0.1080	0.6330	5353	83,870,000
Al Mg4.5Mn, Cld RI ^{c,e,f}	4, 12	298	363	476	(13)	1045	-0.1372	1045	-0.1394	0.6681	2800	2,000,000
2024-T4 Al ^{c,f}	13, 14	303	476	631	35	1944	-0.1755	2499	-0.1969	0.5031	5100	1,836,000
2024-T3 Al ^{c,e,f}	4	378	486	578	(17.3)	1328	-0.1324	1611	-0.1529	0.5655	4249	2,700,000
2014-T6 Al ^e	15	438	494	581	(13.6)	1120	-0.1221	949	-0.1080	0.4803	1200	7,260,000
2024-T3 Al ^e	5, 6	359	497	591	(20.3)	1602	-0.1544	1772	-0.1626	0.4596	663	1,404,000
2024-T4 Al ^e	15	335	502	666	(21.4)	970	-0.1051	897	-0.09771	0.4090	1000	29,400,000
7075-T6 Al ^e	15	489	567	729	(16.5)	4402	-0.2624	3378	-0.2412	0.4150	1000	929,000
7075-T6 Al ^e	5, 6	521	572	736	(12.3)	4324	-0.2388	4546	-0.2419	0.4774	5,182	1,992,000
Ti-6Al-4V ^{c,e}	4	1006	1034	1271	(14.5)	2262	-0.1285	2452	-0.1379	0.5431	214	1,100,000

^aAbbreviations: St = steel, SS = stainless steel, Al = aluminium alloy, HB = Brinell hardness, HRC = Rockwell hardness, El Pol = electropolished, Soln An = solution annealed, H900, etc. = heat treatment conditions, Norm = normalized, Hard = hardened, Cld RI = cold rolled.
^bNumbers refer to the list of References at the end of the paper.
^cSamples initially overstrained before testing.
^dTrue fracture strengths estimated from $\sigma_u + 345$ MPa.
^eTrue fracture strengths estimated by scaling data on similar material.
^fShort life data omitted in stress-life fits; for the $\sigma_m = 0$ fits, the shortest life data points employed were $N_f = 4750$ cycles for Al Mg4.5Mn, 4200 cycles for 2024-T4 Al, and 2050 cycles for 2024-T3 Al.
^gReduction in area is listed, or if not available, elongation is listed.

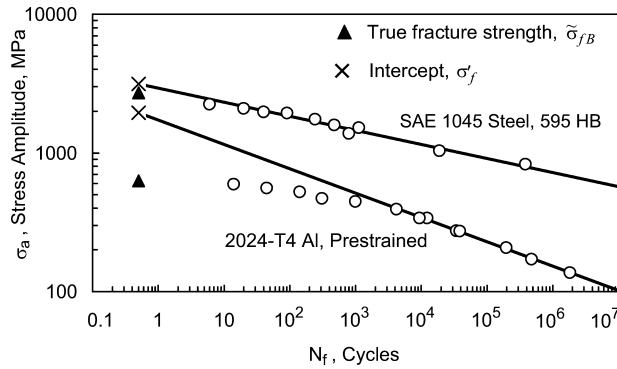


Fig. 2 Stress-life curves for a steel and for an aluminium alloy for zero mean stress, showing intercepts and true fracture strengths.

lives, with the lower line in Fig. 2 being an example of this behaviour. For these three cases, which are identified by a footnote in Table 1, the constants given correspond to a fit that omits the short life data that depart from the trend of the remaining data.

To obtain the estimated fatigue life N_f for cases of non-zero mean stress, an additional equation, $\sigma_{ar} = f(\sigma_a, \sigma_m)$, is needed to calculate σ_{ar} for use with Eq. (2). In this context, σ_{ar} can be thought of as an *equivalent completely reversed stress amplitude*. Below, various $\sigma_{ar} = f(\sigma_a, \sigma_m)$ equations are presented, as is a special data fitting procedure for the Walker method.

Equivalent completely reversed stress amplitude equations

The traditional Goodman relationship,¹⁶ as employed by J.O. Smith,¹⁷ includes the ultimate tensile strength, σ_u .

$$\sigma_{ar} = \frac{\sigma_a}{1 - \sigma_m/\sigma_u}. \quad (3)$$

The equation of Morrow¹⁸ has the same form but employs the true fracture strength $\tilde{\sigma}_{fB}$, as corrected according to Bridgman,¹⁹ in place of σ_u . An alternate form that is often used is obtained by substituting the fitting constant σ'_f , so that the two forms are

$$\sigma_{ar} = \frac{\sigma_a}{1 - \sigma_m/\tilde{\sigma}_{fB}}, \quad \sigma_{ar} = \frac{\sigma_a}{1 - \sigma_m/\sigma'_f}. \quad (a, b) \quad (4)$$

For Eq. (2), note that σ'_f is the stress intercept at $N_f = 0.5$ cycles. Interpreting a tension test as a fatigue test with this life implies the estimate $\sigma'_f = \tilde{\sigma}_{fB}$, which is the logic leading to Eq. (4b). The $\sigma'_f = \tilde{\sigma}_{fB}$ estimate is often quite good for steels, as shown by Landgraf,²⁰ but may be very inaccurate for aluminium alloys, as for the cases illustrated in Fig. 2. Equation (4a) has the disadvantage that true fracture strengths $\tilde{\sigma}_{fB}$ are often not available and then must be estimated.

Another relationship is that of Smith, Watson and Topper.³ Three equivalent forms, related by manipulations employing Eq. (1), are

$$\begin{aligned} \sigma_{ar} &= \sqrt{\sigma_{\max} \sigma_a} & (a) \\ \sigma_{ar} &= \sigma_{\max} \sqrt{\frac{1-R}{2}} & (b) \\ \sigma_{ar} &= \sigma_a \sqrt{\frac{2}{1-R}} & (c). \end{aligned} \quad (5)$$

For brevity, this method will be termed the SWT method. It has the advantage of simplicity and of not depending on any material constant.

A somewhat similar approach is that of Walker,²¹ which involves an additional materials property, here denoted γ . Three corresponding equivalent forms are

$$\begin{aligned} \sigma_{ar} &= \sigma_{\max}^{1-\gamma} \sigma_a^\gamma & (a) \\ \sigma_{ar} &= \sigma_{\max} \left(\frac{1-R}{2} \right)^\gamma & (b) \\ \sigma_{ar} &= \sigma_a \left(\frac{2}{1-R} \right)^{1-\gamma} & (c). \end{aligned} \quad (6)$$

Note that $\sigma_{\max} = \sigma_m + \sigma_a$, so that Eqs (5) and (6) also provide a relationship $\sigma_{ar} = f(\sigma_a, \sigma_m)$.

Values of γ for Eq. (6) can conceivably vary from 0 to 1, but for metals are usually in the range 0.4 to 0.8. If $\gamma = 0.5$, then Eq. (6) is seen to reduce to Eq. (5). The quantity $(1-\gamma)$ can be thought of as a measure of the material's sensitivity to mean stress. Hence, a metal with $\gamma = 0.4$ is quite sensitive to mean stress, and one with $\gamma = 0.8$ is relatively insensitive.

Another equation that is sometimes employed is the Gerber parabola.

$$\sigma_{ar} = \frac{\sigma_a}{1 - (\sigma_m/\sigma_u)^2}. \quad (7)$$

From the historical review of Mann,²² this method dates to 1874 and is perhaps the oldest mean stress equation. However, Eq. (7) was not included in this study, as it is generally inaccurate and moreover usually non-conservative for tensile mean stresses. Also, it incorrectly predicts a harmful effect of compressive mean stress, and so would have to be employed in some modified form for such cases. These trends can be seen in Fig. 3, where the parabola from Eq. (7) is compared with test data for an alloy steel and with the straight lines from Eqs (3) and (4). (See Table 2 for an explanation of conservative and non-conservative trends.)

Numerous additional equations for estimating mean stress effects exist. For example, some others are discussed

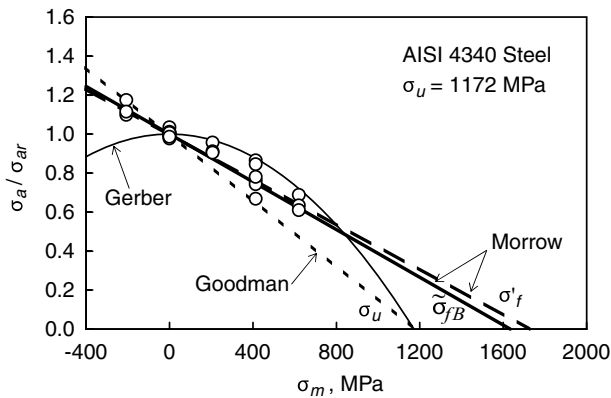


Fig. 3 Normalized stress amplitude–mean plot for AISI 4340 steel, showing the curves for various mean stress equations.

by Nihei,²³ by Chu²⁴ and by Wang.²⁵ The methods of Eqs (3)–(6) were judged by the authors to possess various combinations of wide acceptance, simplicity, and promise as to make them particularly worthy of study.

Fitting the Walker equation

In applying any one of the Goodman, Morrow or SWT relationships to make a life estimate, note that the σ_{ar} value from one of Eqs (3) to (5) is employed with Eq. (2), which is based on fitting only the test data for completely reversed ($\sigma_m = 0$) loading. In contrast, the Walker relationship provides an opportunity to fit fatigue life data at various mean stresses all together, with γ being obtained as part of the fitting process. The procedure involves a multiple linear regression. Details follow.

To proceed, it is convenient to temporarily employ an alternate form of Eq. (2).

$$\sigma_{ar} = AN_f^b, \quad \text{where} \quad A = \sigma'_f 2^b. \quad (8)$$

Combine this with Eq. (6b) and solve for the dependent variable N_f .

$$\sigma_{ar} = AN_f^b = \sigma_{\max} \left(\frac{1-R}{2} \right)^\gamma \quad (a)$$

$$N_f = \left[\sigma_{\max} \left(\frac{1-R}{2} \right)^\gamma \frac{1}{A} \right]^{1/b} \quad (b). \quad (9)$$

Then take the logarithm with base 10 of both sides.

$$\log N_f = \frac{1}{b} \log \sigma_{\max} + \frac{\gamma}{b} \log \left(\frac{1-R}{2} \right) - \frac{1}{b} \log A. \quad (10)$$

A multiple linear regression can then be performed.

$$y = m_1 x_1 + m_2 x_2 + d, \quad (11)$$

where

$$y = \log N_f, \quad x_1 = \log \sigma_{\max}, \quad x_2 = \log \left(\frac{1-R}{2} \right) \quad (12)$$

$$m_1 = \frac{1}{b}, \quad m_2 = \frac{\gamma}{b}, \quad d = -\frac{1}{b} \log A.$$

The regression yields values of the fitting constants m_1 , m_2 and d , which in turn can be used to calculate the desired constants σ'_f , b and γ .

$$b = \frac{1}{m_1}, \quad \gamma = b m_2 = \frac{m_2}{m_1} \quad (13)$$

$$A = 10^{-db} = 10^{-d/m_1}, \quad \sigma'_f = \frac{A}{2^b}.$$

To distinguish values of σ'_f and b obtained by this procedure from those obtained by mere fitting of zero mean stress data, subscripts w for Walker are added, so that the resulting three constants are denoted σ'_{fw} , b_w and γ . Hence, the fitted stress-life curve is

$$\sigma_{ar} = \sigma'_{fw} (2 N_f)^{b_w} \quad (14)$$

with σ_{ar} given by Eq. (6) and the fitted γ . Table 1 gives the resulting constants from fits of this type for each data set.

For the three aluminium alloys having behaviour similar to the lower curve in Fig. 2, a number of cycles N_2 was chosen by judgment as the separation between the long and short life behaviour. The Eq. (14) fit was applied only to the data with $N_f > N_2$, which in these cases included all of the data with mean stresses that differed significantly from zero.

Table 2 Explanation of trends in graphical comparisons

Trend seen on a normalized amplitude–mean plot σ_a / σ_{ar} vs. σ_m or σ_a / σ_{ar} vs. σ_m / σ_{ar}	Trend seen on an equivalent completely reversed stress amplitude versus life plot σ_{ar} vs. N_f	Consequence for life estimates
Data above line or curve for a given model	Data above $\sigma_m = 0$ ($R = -1$) line, or above Walker fit line	Conservative: Actual fatigue life will be longer than estimated
Data below line or curve for a given model	Data below $\sigma_m = 0$ ($R = -1$) line, or below Walker fit line	Non-conservative: Actual fatigue life will be shorter than estimated

COMPARISONS WITH TEST DATA

Fatigue test data for various mean stresses can be compared with candidate mean stress equations $\sigma_{ar} = f(\sigma_a, \sigma_m)$ on stress amplitude versus mean plots, as in Fig. 3. However, a better sense of the success of such an equation for estimating fatigue lives is provided by plotting σ_{ar} values versus fatigue lives N_f . All of the data sets were compared with Eqs (3)–(6) by both of these graphical methods. In what follows in this section, the two graphical methods are described in more detail and examples are presented, and some comments are also given on the data sets that were analysed.

Normalized amplitude–mean plots

The relationships of Goodman or Morrow can be compared to test data by plotting normalized stress amplitudes σ_a/σ_{ar} versus the mean stress σ_m . For the Goodman relationship, Eq. (3), a straight line is expected between the points $\sigma_a/\sigma_{ar} = 1$ and $\sigma_m = \sigma_u$. The two forms of the Morrow relationship, Eqs (4a) and (4b), give similar lines, but with intercepts at $\sigma_m = \tilde{\sigma}_{fB}$ and $\sigma_m = \sigma'_f$, respectively. Example comparisons with test data on this basis are shown for steels in Figs 3 and 4 and for an aluminium alloy in Fig. 5. For the data points shown, each σ_a and σ_m combination corresponds to a laboratory test, and in each case the σ_{ar} value used to normalize σ_a is determined from the experimental life N_f using Eq. (2).

For the steels, note that the data are above the Goodman line for tensile mean stresses, that is, higher stresses than predicted by this method are required to cause fatigue failure. Hence, fatigue life predictions made from a Goodman σ_{ar} value for tensile mean stresses will be shorter than actual fatigue lives, and the method thus errs on the conservative side. (See Table 2.) However, for compressive mean stresses, the opposite is true, and the method is non-conservative. Either form of the Morrow relationship is

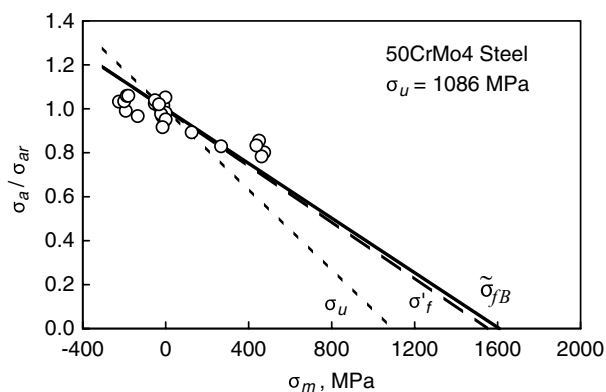


Fig. 4 Normalized stress amplitude–mean plot for 50CrMo4 steel.

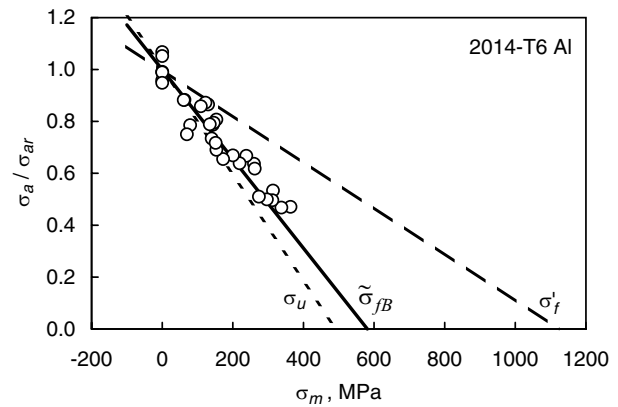


Fig. 5 Normalized stress amplitude–mean plot for 2014-T6 aluminium.

in better agreement with the test data for both tensile and compressive mean stresses. For the aluminium alloy, the Goodman equation is again conservative for tensile mean stresses, and the Morrow relationship of Eq. (4a) with $\tilde{\sigma}_{fB}$ is in good agreement with the data. However, the Morrow relationship of Eq. (4b) with σ'_f disagrees with the data and is non-conservative to a drastic degree. This trend seems to be associated with a stress-life curve that flattens at short life, as for the aluminium alloy in Fig. 2.

On a plot of σ_a/σ_{ar} versus σ_m , the SWT and Walker relationships produce a family of curves, complicating comparisons with test data. However, both form a single curve on a plot with the mean stress axis also normalized to σ_{ar} , that is, on a plot of σ_a/σ_{ar} versus σ_m/σ_{ar} . Comparisons on this basis are shown in Figs 6–8 for the same three materials, with σ_{ar} for the Walker curves being obtained from Eq. (14). For the steels, the fitted value of γ for the Walker relationship gives much better agreement with the data than $\gamma = 0.5$ corresponding to the SWT

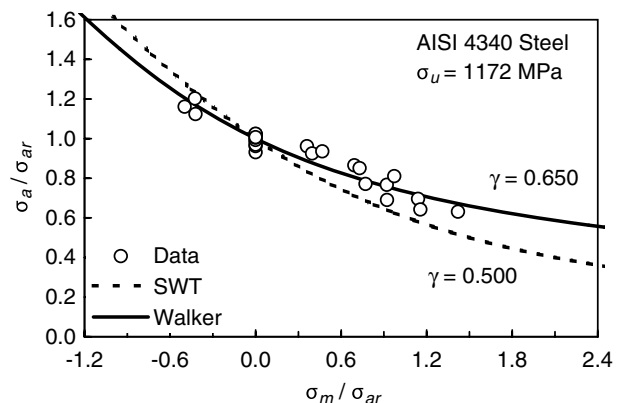


Fig. 6 SWT and Walker amplitude–mean curves for AISI 4340 steel.

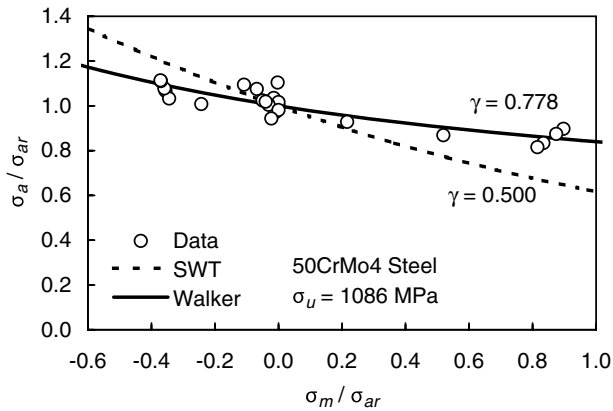


Fig. 7 SWT and Walker amplitude-mean curves for 50CrMo4 steel.

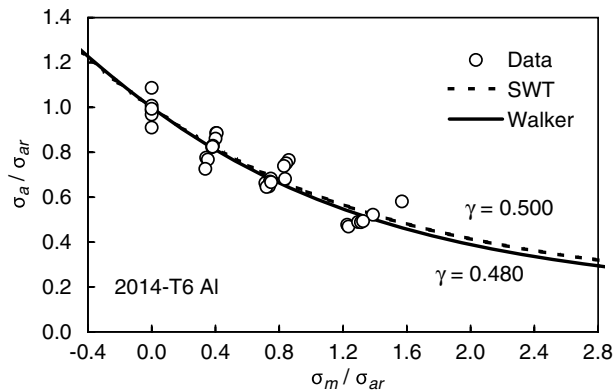


Fig. 8 SWT and Walker amplitude-mean curves for 2014-T6 aluminium.

equation. However, for the aluminium alloy, the fitted γ is close to 0.5, so that both the Walker and SWT methods are in good agreement with the data.

The other steels studied had behaviour generally similar to that of the two steels illustrated, except that the Walker curve may be more shallow, more steep, or similar to the SWT one, depending on the fitted γ value. Further, the other aluminium alloys had behaviour similar to that of the one illustrated, except that the Walker and SWT curves differed considerably for the two lowest strength alloys, for which $\gamma \approx 0.65$.

Life plots

Now consider comparison of mean stress equations and test data on a stress-life plot of σ_{ar} versus N_f . In this case, σ_{ar} is calculated for each test data point using one of the $\sigma_{ar} = f(\sigma_a, \sigma_m)$ expressions above, and these values are plotted versus the experimental fatigue lives N_f . Example plots of this type for 50CrMo4 steel and 2014-T6 aluminium are shown in Figs 9 and 10. Note that for each

material there are four plots, one for each of Eqs (3), (4a), (4b) and (5). On any particular plot, the degree to which the resulting data points consolidate with the line of Eq. (2) gives an indication of the success of the mean stress relationship. (Recall that the Eq. 2 constants are obtained from fitting the $\sigma_m = 0$ data.)

For the Walker relationship, σ_{ar} values from Eq. (6) are similarly plotted versus the N_f data for comparison with the special fit of Eq. (14), which is noted to result from fitting the full set of data at all mean stresses. Comparisons of this type are shown in Figs 11 and 12 for 50CrMo4 steel and 2014-T6 aluminium.

The trends seen in the σ_{ar} versus N_f plots correspond to those already noted for steels and aluminium alloys on the two types of amplitude-mean plot. On any particular plot, where test data points lie above the line for zero mean stress, the estimated lives are too short, and the method errs on the conservative side. Conversely, data points below the line indicate that the method is non-conservative. (See Table 2.) The σ_{ar} versus N_f comparisons shown for 50CrMo4 steel and 2014-T6 aluminium are typical of the steels and aluminium alloys studied, respectively. Where the Walker γ differs significantly from 0.5, the Walker correlation is noticeably better than the SWT one, and the Walker method always gives very good correlation of the test data.

Comments on the data sets analysed

For 18 of the 28 data sets, the tests were conducted in strain control, with stress control sometimes being substituted for tests at relatively long lives where the strains were fully elastic. However, only stress control was employed for the seven data sets referenced to sources in the 1950s, and also for the three data sets obtained from MIL-HDBK-5H.⁹

Tabulated numerical data were utilized, usually from the source referenced, but in a few cases from data obtained through contacting the original authors. The only exceptions were the data from MIL-HDBK-5H, which were digitized from stress-life plots in that publication.

In strain-controlled tests at very short lives, that is, around $N_f < 50$ cycles, and especially for $N_f < 10$ cycles, the very large strains involved may cause buckling or other anomalous behaviour that affects the life. Hence, data points in this range that seemed to depart from the trend of the data at longer lives were not included, specifically, by omitting all points with N_f less than a value chosen as a matter of judgment. In the data sets for stress-controlled tests, there were sometimes points with anomalously short lives where the maximum stress was around or even greater than the material's yield strength. Such data points were omitted, as they are surely affected by large amounts cycle-dependent creep (ratcheting).

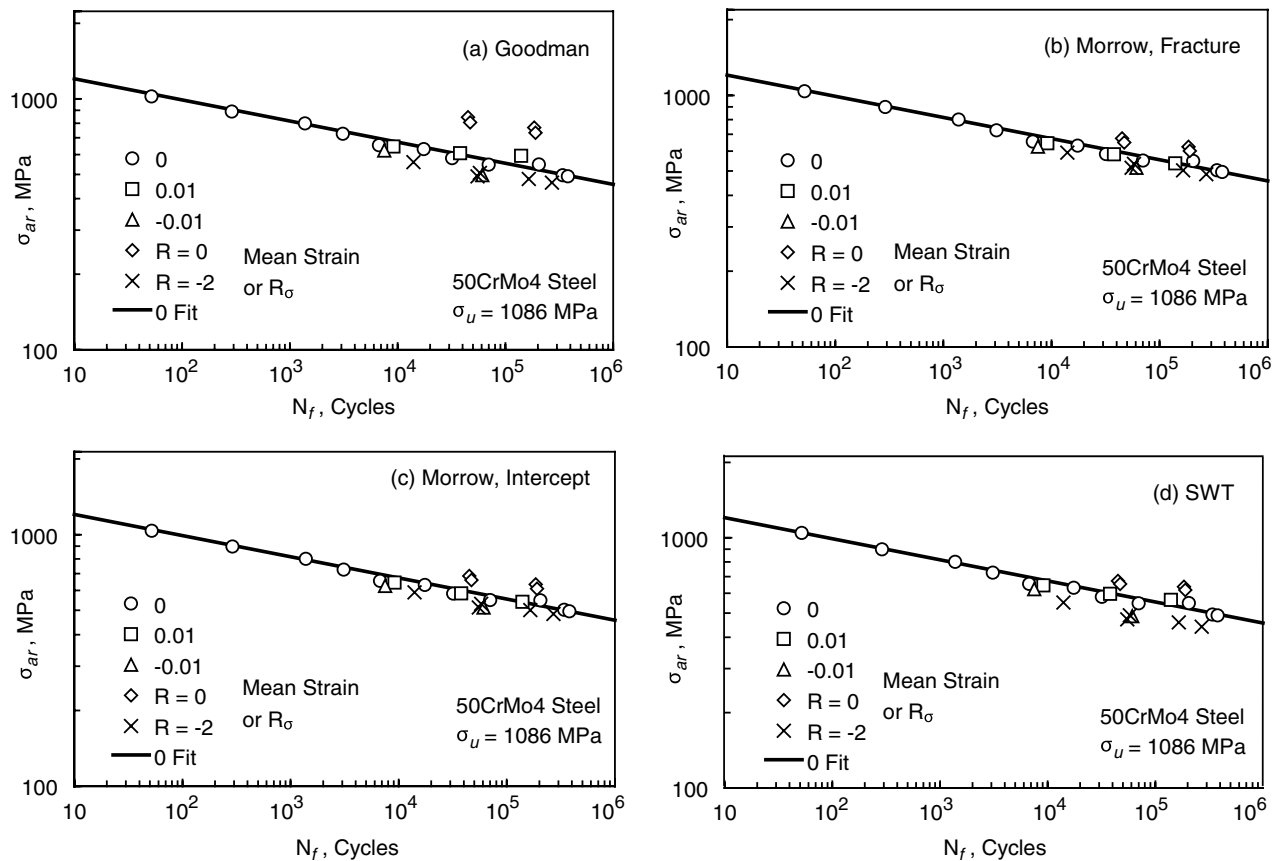


Fig. 9 Equivalent completely reversed stress amplitude versus life correlations for 50CrMo4 steel for the (a) Goodman, (b) Morrow fracture strength, (c) Morrow intercept, and (d) SWT methods.

Many of the data sets had a small number of points at lives around or beyond $N_f = 10^6$ cycles that departed from the trend of the data at shorter lives. Also, there were sometimes a few tests stopped without failure (runouts), where the material began to exhibit fatigue limit behaviour. There was seldom sufficient data of these types to enable any sort of systematic analysis, and so no attempt was made to do so. Hence, in such cases, all points were omitted with N_f greater than a value chosen as a matter of judgment. As a result of these omissions, the data points included for steels exceed $N_f = 2 \times 10^6$ cycles in only one case, but the data included for some aluminium alloys extend well beyond this. Table 1 gives the smallest and largest N_f values that are included in the analysis for each data set. As indicated by a note to the table for three aluminium alloys, short-life data points were excluded, so that anomalous behaviour similar to that of Fig. 2 is not included in the analysis. For these, the smallest N_f values included in the Walker fits of Eq. (14) are the values tabulated, and the smallest N_f values for the $\sigma_m = 0$ fits of Eq. (2) are given in the note to the table.

For some data sets, the samples were subjected to several cycles of initial overstrain at a level around ± 1 to

$\pm 2\%$ strain. This follows the logic of Topper and co-workers,^{14,26} whose work indicates that such overstrains are useful in advancing the fatigue damage process, so that the behaviour is similar to that during service loading with occasional severe cycles. These cases are identified by a footnote in Table 1.

Additional data sets were located, as in MIL-HDBK-5H, but were not included, as they covered only a narrow range of N_f values or included only a small amount of data for non-zero mean stresses. The data on hot-rolled and normalized SAE 1045 steel of Fatemi²⁷ were analysed, but the results are not reported here. Relaxation during these strain-controlled tests resulted in quite small mean stress values, which hardly exceeded the scatter in the zero mean stress data, causing difficulty with the Walker fit. As reported by Fatemi, this set of data is correlated well by the SWT equation. Also analysed but not included are the data on Ti-6Al-4V generated in a U.S. Air Force program.²⁸ In this extensive data set, the stress-life curve begins to flatten around $N_f = 3 \times 10^5$ cycles and the data extend to 10^7 cycles. It was omitted only because a more complex form of stress-life equation would be needed than for the other data

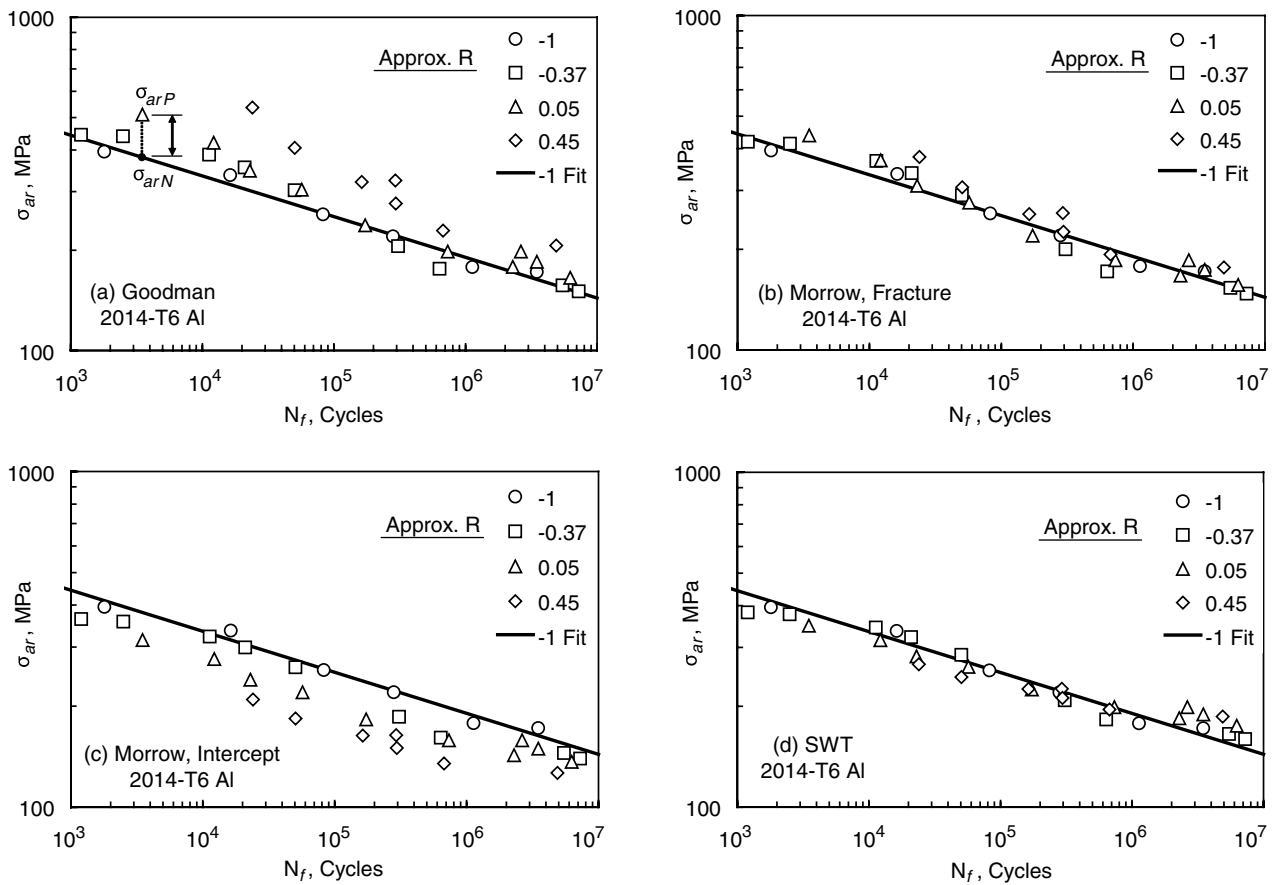


Fig. 10 Equivalent completely reversed stress amplitude versus life correlations for 2014-T6 aluminium for the (a) Goodman, (b) Morrow fracture strength, (c) Morrow intercept and (d) SWT methods.

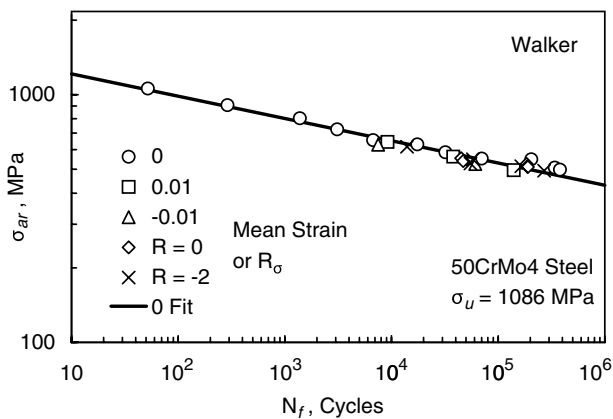


Fig. 11 Equivalent completely reversed stress amplitude versus life for the Walker method applied to 50CrMo4 steel, for which $\gamma = 0.778$.

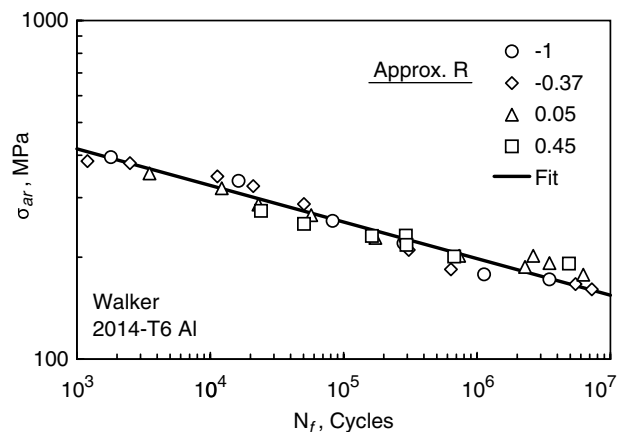


Fig. 12 Equivalent completely reversed stress amplitude versus life for the Walker method applied to 2014-T6 aluminium, for which $\gamma = 0.480$.

sets. However, the SWT equation provides an excellent correlation.

In applying the Morrow relationship in the form of Eq. (4a), it was necessary in many cases to estimate the true

fracture strength $\bar{\sigma}_{FB}$, as this was not given with the data. For most of the aluminium alloys and for the titanium alloy, $\bar{\sigma}_{FB}$ values for a different batch of the same or a very similar alloy were found in the 1998 SAE Handbook²⁹

or in Smith *et al.*³⁰ The missing values were then estimated by multiplying the known one by the ratio of the ultimate tensile strengths σ_u for the two batches of material.

$$\tilde{\sigma}_{fB2} = \tilde{\sigma}_{fB1} \frac{\sigma_{u2}}{\sigma_{u1}}. \quad (15)$$

However, for 6061-T6 aluminium, no value could be found in the literature, and so one was obtained by conducting three tension tests on this alloy, averaging the resulting properties, and then applying Eq. (15).

For steels where $\tilde{\sigma}_{fB}$ was not given with the data, the following estimate was employed:

$$\tilde{\sigma}_{fB} = \sigma_u + 345 \text{ MPa}. \quad (16)$$

This estimate has been widely employed elsewhere but is commonly considered to be valid only for σ_u up to about 1700 MPa. However, the Eq. (16) estimate was compared with $\tilde{\sigma}_{fB}$ data for 25 steels with σ_u fairly evenly spread over the range 400 to 2600 MPa, and it was found to provide a satisfactory correlation over this entire range. The data examined include those in Table 1 of this study where no estimate was necessary, as well as data in Landgraf¹⁰ and in Tables 4.2, 4.6 and 14.1 of the book by Dowling.³¹

DISCUSSION

Graphical comparisons as in Figs 3–12 provide a good sense of the success of the various mean stress equations under study. However, it is not feasible to present similar plots for all 28 sets of data examined, and even if this were done, the information provided would be in very diffuse form. Hence, some means of quantitatively summarizing the success of a given σ_{ar} equation for a given set of experimental data is needed. In this section of the paper, we will first describe such a method, which will then serve as a basis for evaluating and discussing the relative success of the mean stress relationships of Eqs (3)–(6). Next, we will examine trends in values of the fitting constant γ of the Walker equation, and, finally, we will briefly consider extensions of mean stress equations to strain-life curves.

Quantifying comparisons of mean stress equations and test data

Consider σ_{ar} versus N_f plots as in Figs 9 and 10. If the (σ_{ar} , N_f) points for the experimental data for non-zero mean stress all lie close to the zero mean stress fit, corresponding to Eq. (2), then the mean stress equation used to calculate the σ_{ar} values is quite successful. Conversely, the mean stress equation is less successful to the extent that the data scatter widely. To obtain a quantitative measure of the scatter, consider the points σ_{arP} and σ_{arN} as labelled in Fig. 10a. The quantity σ_{arP} corresponds to the value calculated from the stresses applied in the test, plotted

versus the experimental life N_f . And σ_{arN} is the point on the zero mean stress fit at the same life N_f . Hence, σ_{arN} is calculated by entering Eq. (2) with the N_f value corresponding to σ_{arP} .

The difference ($\sigma_{arP} - \sigma_{arN}$) is a measure of the discrepancy, and a dimensionless measure is obtained by normalizing this to the σ_{arN} value.

$$z = \frac{\sigma_{arP} - \sigma_{arN}}{\sigma_{arN}} = \frac{\sigma_{arP}}{\sigma_{arN}} - 1. \quad (17)$$

The standard deviation of the variable z is considered to be a measure of the overall scatter. Considering all non-zero mean stress data points in a given comparison, where n is the number of these, we have

$$s_z = \sqrt{\frac{\sum_{i=1}^n z_i^2}{n-1}}. \quad (18)$$

We will call this s_z the *stress deviation*.

To similarly summarize the success of a Walker equation fit, such as those of Figs 11 and 12, the Eq. (14) line is used as the basis of the comparison. Hence, Eq. (14) is used to compute σ_{arN} for each N_f value. All data points in the range of the Eq. (14) fit are included in the calculation, including the zero mean stress ones.

Success of the various mean stress equations

Values of the *stress deviation* s_z of Eq. (18) were calculated as just described for all 28 data sets, in each case for σ_{ar} values calculated from each of Eqs (3), (4a, b), (5) and (6). These s_z values are given in Table 3, as are the numbers of data points employed in this analysis for each material, both the total numbers and the numbers with approximately zero mean stress. Also, the s_z values are shown in bar chart form for steels in Fig. 13 and for aluminium alloys and the single titanium alloy in Fig. 14. In each bar chart, the materials are ordered according to the ultimate tensile strength σ_u , and the numbers on the horizontal axes correspond to numbers given in Table 3. Of course, smaller s_z values indicate greater success and larger ones the opposite. In Table 3 and Fig. 13, a comparison is also shown for estimated γ values for steels, which will be discussed later.

Plots similar to Figs 3–12 for all data sets, as summarized by the bar charts of Figs 13 and 14 and the values in Table 3, lead to the following observations concerning the success of the various mean stress equations:

The Goodman relationship of Eq. (3), which employs the ultimate tensile strength σ_u , is quite inaccurate and is considerably less accurate than any of the other equations examined. Values of s_z for steels are generally around 0.2, and for aluminium alloys generally around 0.3 or above. The method is excessively conservative for tensile mean stresses, but non-conservative for compressive ones.

Table 3 Values of stress deviation for each material and mean stress equation

Material	Source ^a	Number on bar chart ^b	Number of data points		s_z , Stress deviation					
			Total	$\sigma_m \approx 0$	Goodman	Morrow $\tilde{\sigma}_{fB}$	Morrow σ'_f	SWT	Walker Fit γ	Walker Est. γ
SAE 1015 St	2, 3	1	32	12	0.142	0.080	0.075	0.118	0.069	0.070
GSMnNi63 St	4	2	31	14	0.205	0.129	0.147	0.195	0.116	0.133
Ck45 St	4	3	33	21	0.098	0.082	0.082	0.094	0.069	0.080
SAE 4130 St, Norm	5, 6	4	65	19	0.382	0.217	0.211	0.220	0.107	0.134
49MnVS3 St	4	5	30	16	0.129	0.108	0.102	0.126	0.061	0.097
17MnCrMo33 St	4	6	70	23	0.269	0.079	0.109	0.077	0.061	0.093
CC 450 SS, H1150	7	7	25	11	0.069	0.044	0.044	0.038	0.042	0.040
CC 450 SS, Soln An	7	8	25	9	0.224	0.124	0.139	0.086	0.111	0.080
50CrMo4 St, El Pol	4	9	25	11	0.248	0.104	0.114	0.143	0.041	0.081
42CrMo4 St	4	10	29	19	0.237	0.120	0.133	0.146	0.047	0.082
AISI 4340 St	8	11	25	10	0.236	0.077	0.070	0.118	0.048	0.062
SAE 4130 St, Hard	5, 6	12	33	18	0.170	0.111	0.111	0.096	0.103	0.096
CC 450 SS, H900	7	13	28	9	0.159	0.087	0.100	0.076	0.077	0.105
PH13-8Mo SS, H1000	9	14	17	6	0.525	0.251	0.061	0.108	0.046	0.067
300M St	9	15	71	27	0.178	0.145	0.233	0.123	0.099	0.119
SAE 1045 St, 705 HB	10	16	13	8	0.091	0.083	0.055	0.059	0.052	0.060
SAE 1045 St, 55 HRC	11	17	28	13	0.139	0.096	0.148	0.084	0.086	0.069
SAE 1045 St, 595 HB	10	18	16	9	0.090	0.068	0.084	0.105	0.070	0.120
Steel Average					0.200	0.111	0.112	0.112	0.073	0.088
6061-T6 Al	9	1	50	13	0.518	0.179	0.147	0.222	0.117	–
Al Mg4.5Mn, Cld Rl	4, 12	2	22	11	0.655	0.218	0.191	0.188	0.078	–
2024-T4 Al	13, 14	3	45	7	0.205	0.130	0.233	0.061	0.052	–
2024-T3 Al	4	4	22	6	0.344	0.120	0.266	0.124	0.105	–
2014-T6 Al	15	5	32	6	0.265	0.101	0.199	0.091	0.066	–
2024-T3 Al	5, 6	6	68	14	0.281	0.118	0.252	0.083	0.081	–
2024-T4 Al	15	7	37	12	0.467	0.132	0.249	0.107	0.067	–
7075-T6 Al	15	8	34	9	0.243	0.203	0.394	0.150	0.102	–
7075-T6 Al	5, 6	9	112	49	0.439	0.172	0.327	0.165	0.156	–
Aluminum Average					0.380	0.152	0.251	0.132	0.091	–
Ti-6Al-4V	4	10	21	14	0.283	0.049	0.267	0.096	0.064	–

^aNumbers refer to the list of References at the end of the paper.^bNumbers identify materials in Fig. 13 or 14.

Amplitude–mean plots, such as those of Figs 3–5, show the source of these trends, as the data are above the Goodman line for tensile mean stresses, but below for compressive ones. (See Table 2.) Consequently, for life plots as in Figs 9a and 10a, the σ_{ar} data for tensile mean stress are often far above the zero mean stress line, and those for compressive mean stress are usually below.

The Morrow relationship in the form of Eq. (4a), with the true fracture strength $\tilde{\sigma}_{fB}$, is considerably more accurate than the Goodman relationship. Values of s_z for steels average around 0.11, and for aluminium alloys around 0.15, which still indicates reasonable agreement, but not as good as for steels. On amplitude–mean plots, such as in Figs 3–5, the corresponding line tends to agree with the data. And for σ_{ar} versus N_f plots as in Figs 9b and 10b, only modest scatter about the zero mean stress line is seen.

For steels, the Morrow relationship of Eq. (4b), with the intercept constant σ'_f , is also reasonably accurate, which is expected, as $\tilde{\sigma}_{fB}$ and σ'_f have similar values for most steels. Values of s_z again average around 0.11, and good correlation can be seen on amplitude–mean plots, such as Figs 3 and 4, as well as on σ_{ar} versus N_f plots, as in Fig. 9c.

However, for aluminium alloys, the Morrow σ'_f method is quite inaccurate, with an average s_z of 0.25. This situation is associated with a stress–life trend similar to that for the aluminium alloy in Fig. 2, where the stress values at short lives tend to approach $\tilde{\sigma}_{fB}$, rather than the considerably larger σ'_f value obtained from fitting data at relatively long lives. The artificially large σ'_f value causes a drastic lack of correlation that can be seen both on amplitude–mean plots, as in Fig. 5, and on σ_{ar} versus N_f plots, as in Fig. 10c. Note that the direction and magnitude of

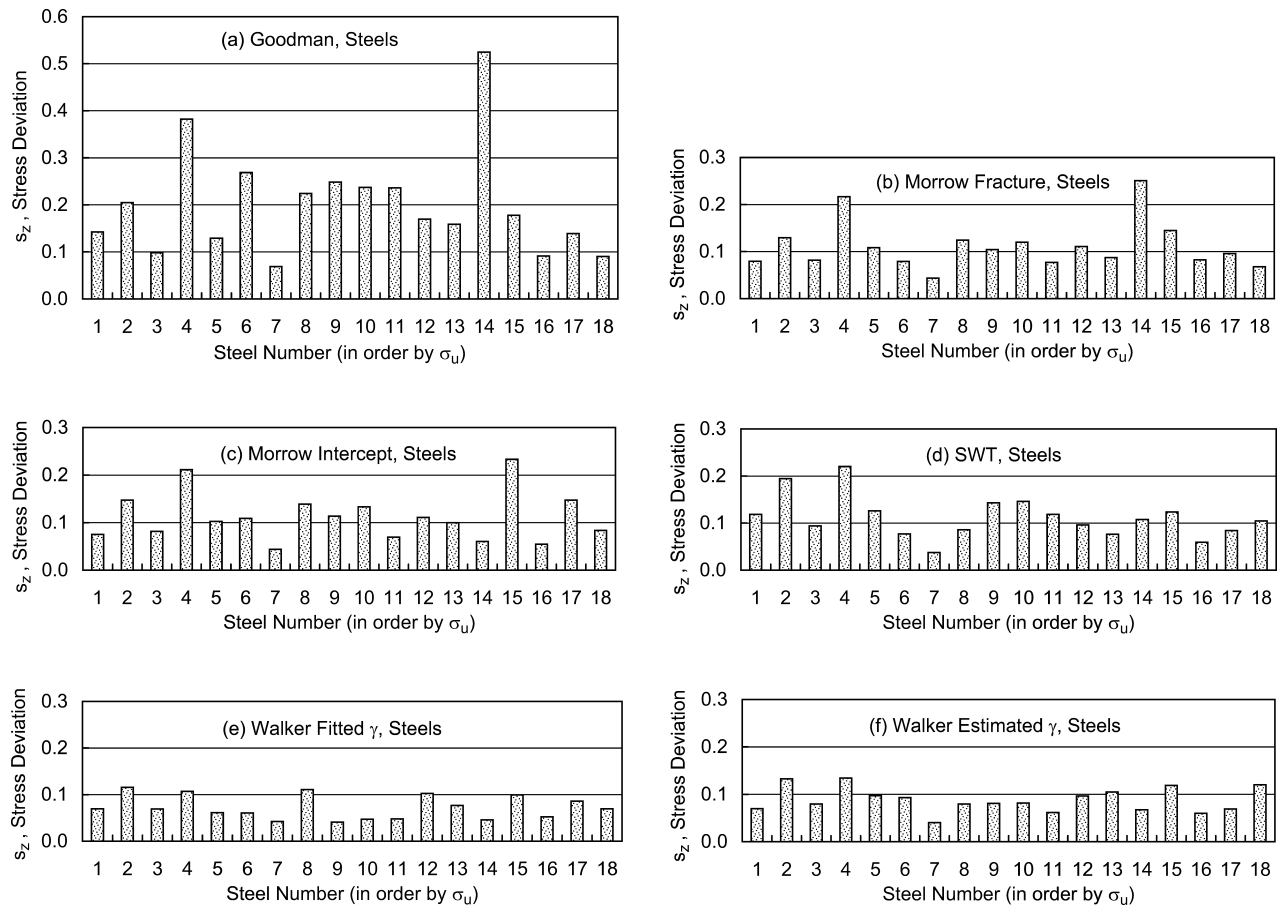


Fig. 13 Stress deviation values s_z for steels for the (a) Goodman, (b) Morrow fracture strength, (c) Morrow intercept, (d) SWT and (e) Walker methods. Also, s_z values are given in (f) for Walker γ values estimated from ultimate tensile strengths. The numbers on the horizontal axes identify the steels listed in Table 3.

the disagreement is such as to cause non-conservative life estimates that can be dangerously inaccurate. Because it is often the case that only relatively long life data are available for fitting Eq. (2), this difficulty with the Morrow σ'_f method represents a fatal flaw as to its usefulness for aluminium alloys.

Considering the SWT method, the overall agreement for steels is similar to that for either version of the Morrow method, with the average s_z for all three being around 0.11. For aluminium alloys, the agreement is better than for even the Morrow $\bar{\sigma}_{FB}$ method, with the average s_z now being 0.13. Also, good correlation is seen on σ_{mr} versus N_f plots, as in Figs 9d and 10d. The SWT method does tend to give somewhat non-conservative estimates for compressive mean stresses in steels, which can be seen in Fig. 9d. However, considering its overall performance, the SWT method is a good choice for general use.

The Walker method gives excellent results where data are available for fitting to evaluate the adjustable param-

eter γ . The average s_z values are 0.07 for steels and 0.09 for aluminium alloys. Noting these values and looking at the bar charts of Figs 13 and 14, as well as the typical life correlations of Figs 11 and 12, it is clear that this method gives the best results of all those considered. This might be expected, as the ability to vary a fitting parameter is not present for any of the other methods. On normalized amplitude–mean plots similar to Figs 6–8, adjusting γ is seen to allow varying the resulting curve to fit data in cases where the SWT method is not very accurate. Moreover, on such a plot, data tend to correlate rather tightly about a curve that fits a particular γ value.

Considering the single titanium alloy studied, it is of course not possible to draw general conclusions for titanium alloys from one set of data. Nevertheless, it is notable that the behaviour was generally similar to that for aluminium alloys, with poor correlation for the Goodman and Morrow σ'_f methods, and good or excellent agreement for the remaining three methods.

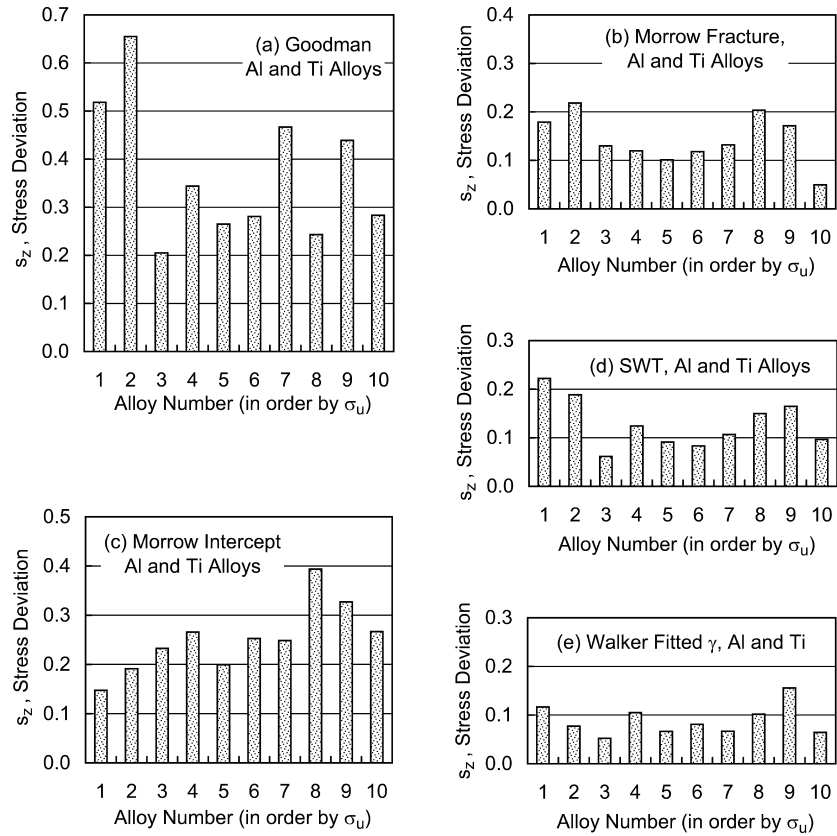


Fig. 14 Stress deviation values s_z for aluminium and titanium alloys for the (a) Goodman, (b) Morrow fracture strength, (c) Morrow intercept, (d) SWT and (e) Walker methods. The numbers on the horizontal axes identify the alloys listed in Table 3.

Comments on errors in life

Consider either an individual z value of 0.10, or a stress deviation s_z for a set of data of 0.10. This corresponds to an error in stress of a factor of 1.10. The error in life associated with this value depends on the exponent b or b_w of Eq. (2) or (14), that is, on the slope of the stress-life curve on a log-log plot. For a typical value of $b = -0.10$, the corresponding error in life, termed the *life factor*, is

$$\text{life factor} = [1 + (z \text{ or } s_z)]^{-1/b} = 1.10^{10} = 2.59 \quad (19)$$

Hence, the fatigue life deviates from the zero mean stress (or Walker-fitted) line by a factor of 2.59. Of course, when the above calculation is applied to a z value, the result characterizes the error in life for only an individual data point, and when applied to an s_z value, the result characterizes the overall error in life for the set of data. For the typical $b = -0.10$, some additional values of the life factor are given below.

z or s_z	0.072	0.10	0.20	0.30	0.50
Life factor	2.00	2.59	6.19	13.8	57.7

Thus, for $b = -0.10$, the z or s_z cannot exceed 0.072 for the life to not be more than a factor of two in error. Values

of s_z similar to the average for the Goodman method applied to steels of 0.2, or the average for aluminium alloys of 0.38, correspond to quite large errors in life, and the same applies for the Morrow σ'_f method applied to aluminium alloys. Also, in the comparisons made for the cases just mentioned, individual data points for the larger mean stress values often had values around $z = 0.5$ or larger, examples of which can be seen in Figs 9a, 10a and 10c. Further, for the same cases, there were s_z values for several sets of data in the range 0.3 to 0.6, which corresponds to quite extreme errors in life.

The values of b and b_w in Table 1 vary considerably, and similar calculations can of course be done for any particular value of b or b_w . For values of $-b$ smaller than 0.10, the life factors increase relative to the values for $-b = 0.10$, and they decrease for $-b$ larger than 0.10. Even considering the variation of b , the overall picture that emerges is that use of the less successful mean stress equations can result in unacceptably large errors in life.

Trends in fitted Walker γ values for steels

The success of the Walker method suggests that it would be worthwhile to develop methods for estimating γ values for materials where non-zero mean stress data for fitting

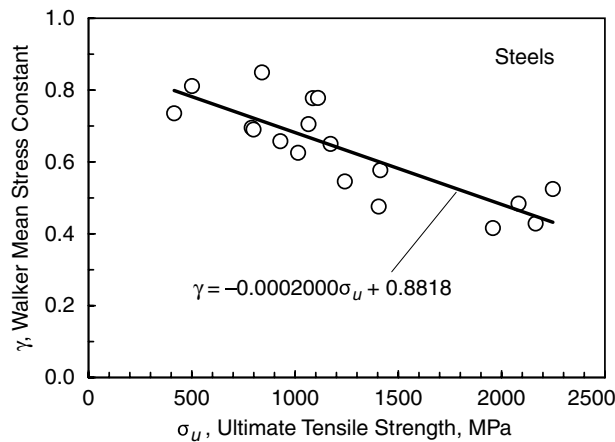


Fig. 15 Trend of fitted γ values with ultimate tensile strength for the Walker method applied to steels.

are not available. We will now explore this possibility for steels and for aluminium alloys.

The fitted Walker γ values from Table 1 for steels are plotted versus the ultimate tensile strength σ_u in Fig. 15. A trend of decreasing γ with increasing σ_u is seen, and fitting a linear relationship gives

$$\gamma = -0.0002000\sigma_u + 0.8818 \quad (\sigma_u \text{ in MPa}). \quad (20)$$

For this fit, the coefficient of determination R^2 is 0.68. A similar trend with essentially the same R^2 also exists between γ and yield strength σ_o . (If desired, the reader can obtain this fit from the data in Table 1.)

The trend of decreasing γ for higher strength steels indicates an increasing sensitivity to mean stress. Also, looking at Eq. (6a), the behaviour of higher strength steels is increasingly determined by the maximum stress $\sigma_{\max}$, and less by the stress amplitude σ_a , indicating more brittle behaviour. This behaviour may be related to the analogous trend seen in the crack growth behaviour of a wide range of materials, as discussed by Ritchie.³²

Equation (20) was employed to estimate the γ value for the steels of this study from their respective σ_u values, and the comparisons with test data were repeated, with one of the life plots being shown in Fig. 16. Because the Eq. (14) fit would not be available where γ needs to be estimated, the σ_{arN} values for these comparisons were calculated from the Eq. (2) fits to the zero mean stress data, and only the non-zero mean-stress data points were employed in the summation of Eq. (18) to obtain s_z . In other words, the same procedure was used as for the evaluations of the non-Walker mean stress methods of Eqs (3)–(5). The bar chart showing all of the resulting stress deviations s_z is given as Fig. 13f, and the values are listed in Table 3. The result is almost as good as for the individually determined γ values used with the Eq. (14) fits. Note that the average s_z increases to 0.088 versus 0.073 for the fitted

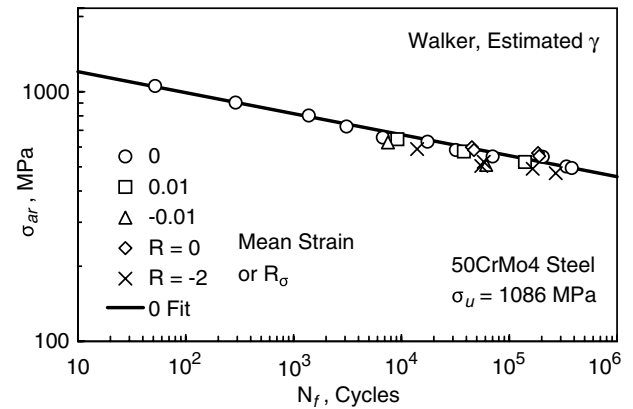


Fig. 16 Equivalent completely reversed stress amplitude versus life for the Walker method applied to 50CrMo4 steel, where an estimated value of $\gamma = 0.665$ is employed.

γ values. This $s_z = 0.088$ value is also somewhat lower than the 0.11 values for the two versions of the Morrow method and the SWT method.

Of course, it is true that the just-described comparisons with data for the estimated γ values involve a degree of circular logic, in that the same data were used to develop the Eq. (20) correlation as to evaluate it. However, the number of 18 data sets for steel is sufficiently large that omitting any one data set and revising Eq. (20) will not significantly affect the comparison for the omitted data set. This contention was tested by omitting, one at a time, several of the data points that fell farthest from the fitted line of Fig. 15, and noting that in each case the fit was only slightly affected.

Trends in fitted Walker γ values for aluminium alloys

Considering aluminium alloys, the fitted γ values are plotted versus ultimate tensile strength σ_u in Fig. 17. The smaller number of data points and their distribution

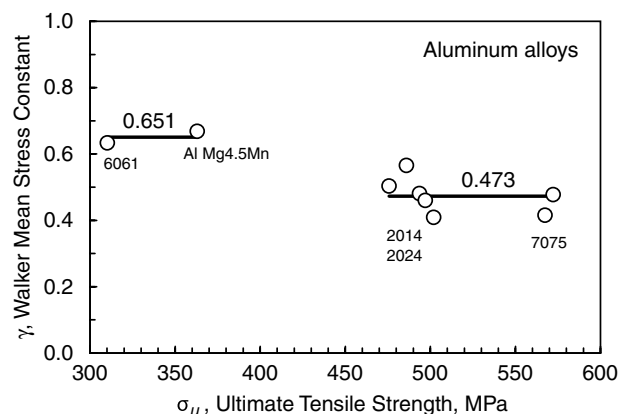


Fig. 17 Trend of fitted γ values with ultimate tensile strength for the Walker method applied to aluminium alloys.

does not seem to justify fitting an overall trend line. Also, the trend with yield strength was examined, but the correlation was poor. The relatively high strength alloys 2014, 2024 and 7075 had values of γ in the range 0.41 to 0.57, scattering around an average of 0.47. This is close enough to $\gamma = 0.50$ that the SWT relationship would appear to suffice for these alloys. The two relatively low strength alloys, 6061-T6 and Al Mg4.5Mn, had values near 0.65, so that this value might be used as an estimate for alloys of similar strength.

As just noted for steels, there appears to be a trend of decreasing γ for higher strength aluminium alloys, although the distribution of the present data makes the trend less clear than for steels.

Extension to strain-life curves

Consider the strain-life curve, which is usually represented by the form

$$\varepsilon_{ar} = \frac{\sigma'_f}{E} (2N_f)^b + \varepsilon'_f (2N_f)^c. \quad (21)$$

Here, the strain amplitude ε_a has the subscript r added to indicate that the curve is obtained from completely reversed ($\sigma_m \approx 0$) test data. Because σ_{ar}/E is the elastic strain amplitude, with E being the elastic modulus, the first term above is consistent with Eq. (2). The second term, with additional fitting constants ε'_f and c , derives from fitting plastic strain amplitudes versus life. Hence, the two terms sum to the total strain amplitude ε_{ar} .

As previously discussed in publication,^{1,31} Eq. (21) can be generalized to include mean stress effects in a manner that is mathematically consistent with any desired mean stress equation, $\sigma_{ar} = f(\sigma_a, \sigma_m)$, such as any of Eqs (3)–(6). In particular, it is found that Eq. (21) becomes

$$\varepsilon_a = \frac{\sigma'_f}{E} (2N^*)^b + \varepsilon'_f (2N^*)^c. \quad (22)$$

The quantity N^* is an equivalent life value corresponding to the zero mean stress case, that is, N^* is simply the life calculated from ε_a as if the mean stress were zero. An expression for N^* can be developed for any particular $\sigma_{ar} = f(\sigma_a, \sigma_m)$.

Because the Walker mean stress relationship has been shown to provide better correlation with test data than other common methods, its incorporation into the strain-life curve is of particular interest. In this case, Eq. (6b) leads to the particular N^* value for the Walker method, here denoted N_w^* .

$$N_w^* = N_f \left(\frac{1-R}{2} \right)^{(1-\gamma)/b} \quad (a)$$

$$N_f = N_w^* \left(\frac{1-R}{2} \right)^{-(1-\gamma)/b} \quad (b). \quad (23)$$

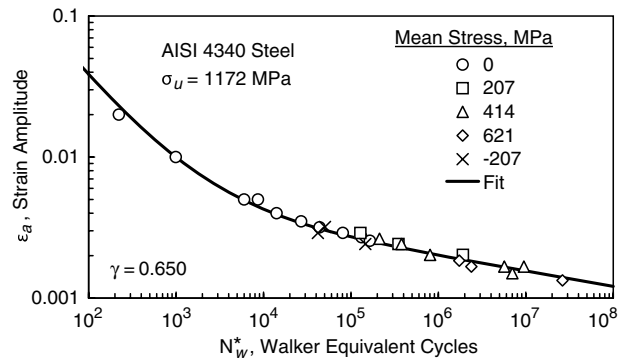


Fig. 18 Strain amplitude versus Walker equivalent life N_w^* for AISI 4340 steel.

Form (23b) is simply (23a) solved for N_f . To calculate the fatigue life, the strain amplitude ε_a is substituted into Eq. (22), and an iterative solution yields N_w^* . Then the life N_f with the mean stress effect included is calculated from Eq. (23b). Also, Eq. (22) indicates that a plot of ε_a versus N_w^* is expected to provide a single curve for all values of mean stress. This is illustrated in Fig. 18 for one of the sets of data with strain-controlled tests at short lives.

An additional journal article is in preparation where the above approach is explored in more detail and is then applied to the 18 data sets of this study that include strain-controlled tests at short lives.

CONCLUSIONS

Based on the analysis and discussion of this paper, the following conclusions can be stated regarding the methods that were examined for accounting for mean stress effects:

- 1 Where life estimates are desired in fatigue analysis, the Goodman method employing the ultimate tensile strength σ_u is highly inaccurate and should not be employed.
- 2 The Morrow equation with the true fracture strength $\tilde{\sigma}_{fB}$ is reasonably accurate in most cases. However, this method has the disadvantage that true fracture strengths $\tilde{\sigma}_{fB}$ are often not available and then must be estimated.
- 3 The Morrow method with the intercept constant σ'_f is also reasonably accurate for steels, but often gives grossly non-conservative life estimates for aluminium alloys.
- 4 The Smith–Watson–Topper (SWT) method provides good results in most cases, and for aluminium alloys it is somewhat more accurate than the Morrow $\tilde{\sigma}_{fB}$ equation. The SWT method has the advantage of simplicity and is a good choice for general use.
- 5 Where data are available for fitting the adjustable parameter γ , the Walker method gives superior results. A straightforward multiple regression procedure allows data at various mean stresses to be fitted all together.

- 6 For steels, the correlation of γ with σ_u of Eq. (20) allows γ to be estimated with sufficient accuracy to still give excellent results for this class of material. Relatively high strength aluminium alloys have $\gamma \approx 0.5$, which corresponds with the SWT method, but higher values of γ apply for relatively low strength aluminium alloys.
- 7 For both steels and aluminium alloys, there is a trend of decreasing γ with increasing strength, indicating an increasing sensitivity to mean stress, and also more brittle behaviour that is increasingly dominated by the maximum stress $\sigma_{\max}$, and less affected by the stress amplitude σ_a .

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